

## Linux on BSD

BSD can run most Linux applications as-is.  
For the rest there are Linux emulators

## TCP/IP

The protocol on which the internet is  
based on was first tested on BSD

installed system. The screens for these functions are straightforward and require no elaboration. Once these steps are completed you are presented with an option to browse through the packages collection.

This will produce a screen listing the many packages that came bundled with the installation disks. As you can see the list is humongous. Of special interest to us are the entries KDE and Gnome. Both are windows managers that offer a much better user experience than the basic X windows system installed as part of the basic install.

The KDE system contains many packages, and rather than selecting each, a quicker way is to select the package titled **kde-3.5.10**. This will automatically select all the other packages as they are dependencies. It needs to be noted that the package selection is not as liberal as the routines found in Linux. So, selectively picking packages within the KDE framework is not allowed. If you try to deselect any superfluous package, say the personal information manager KPIM, a message stating that it is required by another package comes up. It is left to the user to figure out which the other package is. And there is no way to dismiss the warning and proceed at one's risk.

Similarly, the Gnome desktop manager can also be installed. If needed the other packages in the list can be browsed, and selected. Once the relevant packages are selected, the install command can be given. All selected packages are extracted and installed, and when this process is over, the menu can be exited. This brings us to the Post installation configuration menu. Most of the entries in this menu have been covered by the preceding sequence of screens, but there are some tools towards the end that are worth exploring.

The **Configure system ttys** option needs to be used to load the KDE windows manager on reboot. If the installation routine is exited before this step, the installation of the windows managers does not automatically modify the default start up routine.

You need to manually edit the **/etc/ttys** file, which can be accomplished easily with the **Configure System ttys** option. Once selected, after a descriptive message, a command line text editor comes up with the **ttys** file loaded. The screen can befuddle many, so it is best to stick to the basics. Use the down arrow key to scroll down to the line starting with **tttyv8** under the section **# Virtual terminals**. As can be seen, a file "xdm" is being referred to, change the line from:

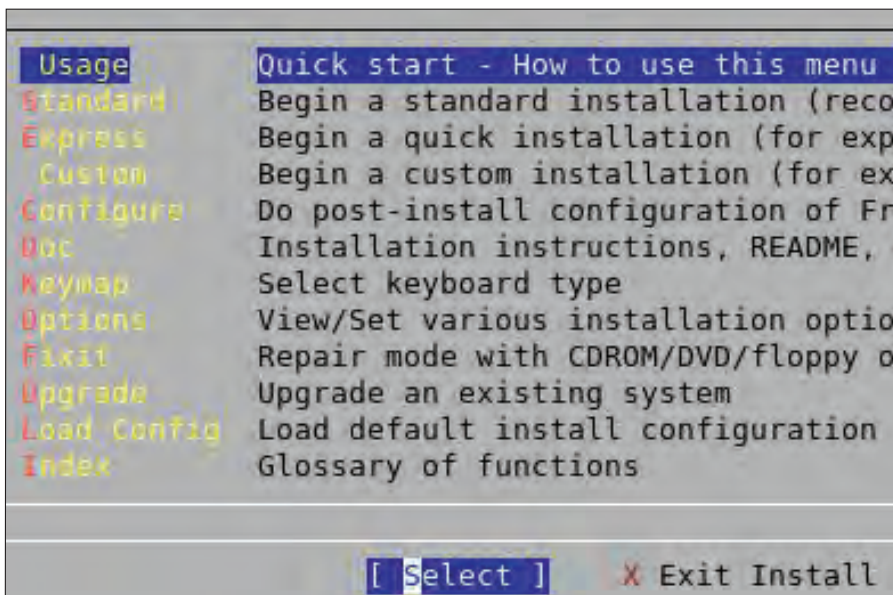
```
tttyv8 "/usr/local/bin/xdm
-nodaemon" xterm off secure
to tttyv8 "/usr/local/bin/kdm
-nodaemon" xterm on secure.
Change the xdm to kdm, and the off
to on. Then save and exit the editor, by
pressing the [Esc] button to bring up
the Menu, select Leave Editor which
will bring up the screen to select Save
Changes. After the editor is closed,
the post install configuration menu is
brought up, which can be exited to reach
the main menu from where the installa-
tion can be terminated. The system will
reboot so the installation media should
be removed before proceeding.
```

On reboot, you will be presented with the welcome sight of the KDM login window. Here you can login using the username and password of the user created previously. And you can select the window manager to be used for the session as well.

## Driver installation

The first challenge is to check if all the devices are detected and relevant drivers installed. The simplest way, without leaving the windows manager, is to use the device—play a song to check audio, try connecting to the Net to check the LAN connection etc. If any test fails, make a note of the device that seems to be non functional. In the case of this writer, the on-board sound was the spoilsport.

To check if a device has been installed, you need the **dmesg** command. This needs to be run as super user, and since a root user cannot log into the system through KDM one cannot issue commands at the terminal inside KDE as well. For all such purposes you need to access one of the other virtual terminals. There are 7 other terminals (tttyv1 to tttyv7) that can be used and each can be accessed directly by using the **[Alt] + [Ctrl] + Fn** key combination. So to access the second virtual terminal use **[Alt] + [Ctrl] + [F2]** combo, and so on. The Windows manager is loaded on the ninth virtual terminal (tttyv8), and can be accessed with the



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